

Math 7243 — Homework 1: Matrix Calculus

All derivatives use the **denominator layout** convention: for $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\frac{\partial f}{\partial \vec{x}}$ is the $n \times 1$ column of partials $\partial f / \partial x_k$; for $\vec{u} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\frac{\partial \vec{u}}{\partial \vec{x}}$ is the $n \times m$ matrix with (k, j) entry $\partial u_j / \partial x_k$.

Problem 1

Assume $\vec{x} \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $\vec{b} \in \mathbb{R}^m$. Let $f(\vec{x}) = \vec{b}^T A \vec{x}$. Find ∇f .

Solution

Let $\vec{a}^T := \vec{b}^T A$, a $1 \times n$ row vector, so that $f(\vec{x}) = \vec{a}^T \vec{x}$. By Problem 6 (or the corresponding lecture proposition), $\nabla f = \vec{a}$. Transposing a product reverses the order, so

$$\nabla f = \vec{a} = (\vec{b}^T A)^T = A^T (\vec{b}^T)^T = A^T \vec{b}.$$

Check. A^T is $n \times m$ and $\vec{b}$ is $m \times 1$, so $A^T \vec{b}$ is $n \times 1$, matching the shape of $\vec{x}$.

Problem 2

Assume $\vec{x} \in \mathbb{R}^n$. Find $\frac{\partial \vec{x}^T \vec{x}}{\partial \vec{x}}$.

Solution

A row vector times a column vector is a scalar:

$$\vec{x}^T \vec{x} = x_1^2 + x_2^2 + \cdots + x_n^2.$$

For each $k = 1, \dots, n$, only the term x_k^2 involves x_k , so

$$\frac{\partial (\vec{x}^T \vec{x})}{\partial x_k} = 2x_k.$$

Stacking these into a column,

$$\frac{\partial (\vec{x}^T \vec{x})}{\partial \vec{x}} = \begin{bmatrix} 2x_1 \\ \vdots \\ 2x_n \end{bmatrix} = 2\vec{x}.$$

Problem 3

Assume $\vec{x}, \vec{a} \in \mathbb{R}^n$. Find $\frac{\partial (\vec{x}^T \vec{a})^2}{\partial \vec{x}}$.

Solution

The quantity $\vec{x}^T \vec{a}$ is a scalar, so $(\vec{x}^T \vec{a})^2$ is the square of a scalar and the chain rule applies. Using $\frac{\partial (\vec{x}^T \vec{a})}{\partial \vec{x}} = \vec{a}$ (Problem 6),

$$\frac{\partial (\vec{x}^T \vec{a})^2}{\partial \vec{x}} = 2(\vec{x}^T \vec{a}) \frac{\partial (\vec{x}^T \vec{a})}{\partial \vec{x}} = 2(\vec{x}^T \vec{a}) \vec{a}.$$

Check. $\vec{x}^T \vec{a}$ is a scalar and $\vec{a}$ is $n \times 1$, so the answer is $n \times 1$. Equivalently, writing $(\vec{x}^T \vec{a})^2 = \vec{x}^T (\vec{a} \vec{a}^T) \vec{x}$ and using the symmetric quadratic-form rule gives $2\vec{a} \vec{a}^T \vec{x}$, the same vector.

Problem 4

Suppose $\vec{x} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ sends $\vec{z} \in \mathbb{R}^n$ to $\vec{x}(\vec{z}) \in \mathbb{R}^m$, suppose $\vec{y} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ likewise, and let A be an $m \times m$ constant matrix. Prove that

$$\frac{\partial (\vec{y}^T A \vec{x})}{\partial \vec{z}} = \frac{\partial \vec{y}}{\partial \vec{z}} A \vec{x} + \frac{\partial \vec{x}}{\partial \vec{z}} A^T \vec{y}.$$

Proof

Write $\vec{y}^T A \vec{x}$ as a double sum over the entries:

$$\vec{y}^T A \vec{x} = \sum_{i=1}^m \sum_{j=1}^m y_i A_{ij} x_j,$$

where each y_i and x_j depends on $\vec{z}$ and each A_{ij} is constant. Fix $k \in \{1, \dots, n\}$ and differentiate with respect to z_k , applying the ordinary product rule to each term:

$$\frac{\partial (\vec{y}^T A \vec{x})}{\partial z_k} = \sum_{i=1}^m \sum_{j=1}^m \frac{\partial y_i}{\partial z_k} A_{ij} x_j + \sum_{i=1}^m \sum_{j=1}^m y_i A_{ij} \frac{\partial x_j}{\partial z_k}.$$

First sum. Summing over j first, $\sum_j A_{ij} x_j = (A \vec{x})_i$, so

$$\sum_{i=1}^m \frac{\partial y_i}{\partial z_k} (A \vec{x})_i = \left(\frac{\partial \vec{y}}{\partial \vec{z}} A \vec{x} \right)_k,$$

since $\partial y_i / \partial z_k$ is the (k, i) entry of $\frac{\partial \vec{y}}{\partial \vec{z}}$.

Second sum. Summing over i first, $\sum_i A_{ij} y_i = (A^T \vec{y})_j$, so

$$\sum_{j=1}^m \frac{\partial x_j}{\partial z_k} (A^T \vec{y})_j = \left(\frac{\partial \vec{x}}{\partial \vec{z}} A^T \vec{y} \right)_k,$$

since $\partial x_j / \partial z_k$ is the (k, j) entry of $\frac{\partial \vec{x}}{\partial \vec{z}}$.

Both sides therefore have the same k -th entry for every $k = 1, \dots, n$, and stacking gives

$$\frac{\partial(\vec{y}^T A \vec{x})}{\partial \vec{z}} = \frac{\partial \vec{y}}{\partial \vec{z}} A \vec{x} + \frac{\partial \vec{x}}{\partial \vec{z}} A^T \vec{y}. \quad \blacksquare$$

Check. Each term is $(n \times m)(m \times m)(m \times 1) = n \times 1$, matching $\vec{z} \in \mathbb{R}^n$.

Problem 5

Suppose $A(x) : \mathbb{R} \rightarrow \mathbb{R}^{n \times n}$. Show that if $A(x)$ is invertible, then $\frac{dA^{-1}}{dx} = -A^{-1} \frac{dA}{dx} A^{-1}$.

Proof

Since $A(x)$ is invertible for each x ,

$$A(x) A^{-1}(x) = I.$$

Differentiate both sides with respect to x . The derivative of a matrix is taken entrywise, the product rule applies with the order of the factors preserved, and I is constant, so its derivative is the zero matrix:

$$\frac{dA}{dx} A^{-1} + A \frac{dA^{-1}}{dx} = 0.$$

Rearranging,

$$A \frac{dA^{-1}}{dx} = -\frac{dA}{dx} A^{-1}.$$

Multiply on the *left* by A^{-1} (matrix multiplication is not commutative, so the side matters), and use $A^{-1}A = I$:

$$\frac{dA^{-1}}{dx} = -A^{-1} \frac{dA}{dx} A^{-1}. \quad \blacksquare$$

Remark. For $n = 1$ this is the familiar rule $\frac{d}{dx} \frac{1}{a(x)} = -\frac{a'(x)}{a(x)^2}$.

Problem 6

Let $\vec{x}, \beta \in \mathbb{R}^p$. Prove that $\frac{\partial \vec{x}^T \beta}{\partial \vec{x}} = \beta$.

Proof

Writing out the product,

$$\vec{x}^T \beta = \sum_{i=1}^p x_i \beta_i.$$

Fix $k \in \{1, \dots, p\}$. Only the term $x_k \beta_k$ involves x_k , and β is constant with respect to $\vec{x}$, so

$$\frac{\partial(\vec{x}^T \beta)}{\partial x_k} = \beta_k.$$

Stacking these partials into a column,

$$\frac{\partial(\vec{x}^T \beta)}{\partial \vec{x}} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} = \beta. \quad \blacksquare$$

Problem 7

(Chain Rule.) Assume Y is an n -vector depending on X , and X depends on some $\vec{z} \in \mathbb{R}^q$; that is, $X : \mathbb{R}^q \rightarrow \mathbb{R}^p$ and $Y : \mathbb{R}^p \rightarrow \mathbb{R}^n$. Show that

$$\frac{\partial Y}{\partial \vec{z}} = \frac{\partial X}{\partial \vec{z}} \frac{\partial Y}{\partial X}.$$

Does the order matter?

Proof

In denominator layout the two factors are

$$\frac{\partial X}{\partial \vec{z}} = \begin{bmatrix} \frac{\partial X_1}{\partial z_1} & \frac{\partial X_2}{\partial z_1} & \dots & \frac{\partial X_p}{\partial z_1} \\ \frac{\partial X_1}{\partial z_2} & \frac{\partial X_2}{\partial z_2} & \dots & \frac{\partial X_p}{\partial z_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial X_1}{\partial z_q} & \frac{\partial X_2}{\partial z_q} & \dots & \frac{\partial X_p}{\partial z_q} \end{bmatrix} \quad (q \times p), \quad \frac{\partial Y}{\partial X} = \begin{bmatrix} \frac{\partial Y_1}{\partial X_1} & \frac{\partial Y_2}{\partial X_1} & \dots & \frac{\partial Y_n}{\partial X_1} \\ \frac{\partial Y_1}{\partial X_2} & \frac{\partial Y_2}{\partial X_2} & \dots & \frac{\partial Y_n}{\partial X_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial Y_1}{\partial X_p} & \frac{\partial Y_2}{\partial X_p} & \dots & \frac{\partial Y_n}{\partial X_p} \end{bmatrix} \quad (p \times n).$$

The (k, j) entry of the product is row k of the first matrix times column j of the second:

$$\left(\frac{\partial X}{\partial \vec{z}} \frac{\partial Y}{\partial X} \right)_{kj} = \frac{\partial X_1}{\partial z_k} \frac{\partial Y_j}{\partial X_1} + \frac{\partial X_2}{\partial z_k} \frac{\partial Y_j}{\partial X_2} + \dots + \frac{\partial X_p}{\partial z_k} \frac{\partial Y_j}{\partial X_p} = \frac{\partial Y_j}{\partial z_k},$$

where the last equality is the multivariable chain rule: z_k influences Y_j only through $X_1, \dots, X_p$. But $\partial Y_j / \partial z_k$ is by definition the (k, j) entry of $\frac{\partial Y}{\partial \vec{z}}$. Since the entries agree for all $k = 1, \dots, q$ and $j = 1, \dots, n$, the two $q \times n$ matrices are equal:

$$\frac{\partial Y}{\partial \vec{z}} = \frac{\partial X}{\partial \vec{z}} \frac{\partial Y}{\partial X}. \quad \blacksquare$$

Does the order matter?

Yes. The sizes are

$$\underbrace{\frac{\partial Y}{\partial \vec{Z}}}_{q \times n} = \underbrace{\frac{\partial X}{\partial \vec{Z}}}_{q \times p} \underbrace{\frac{\partial Y}{\partial X}}_{p \times n},$$

so the stated product is defined and has the correct shape. The reversed product $\frac{\partial Y}{\partial X} \frac{\partial X}{\partial \vec{Z}}$ is $(p \times n)(q \times p)$, which is undefined unless $n = q$, and even then produces a $p \times p$ matrix rather than a $q \times n$ one. The order is forced by the layout convention.

Remark. In numerator layout every matrix above is transposed, and the chain rule instead reads $\frac{\partial Y}{\partial \vec{Z}} = \frac{\partial Y}{\partial X} \frac{\partial X}{\partial \vec{Z}}$.

Problem 8

Let $z : \mathbb{R}^p \rightarrow \mathbb{R}$ depend on $\vec{x} \in \mathbb{R}^p$, and let Y be an n -vector depending on $\vec{x} \in \mathbb{R}^p$. Prove that

$$\frac{\partial}{\partial \vec{x}}(zY) = z \frac{\partial Y}{\partial \vec{x}} + \frac{\partial z}{\partial \vec{x}} Y^T.$$

Proof

Note first that all three expressions are $p \times n$ matrices:

$$\frac{\partial(zY)}{\partial \vec{x}} : p \times n, \quad z \frac{\partial Y}{\partial \vec{x}} : p \times n, \quad \frac{\partial z}{\partial \vec{x}} Y^T : (p \times 1)(1 \times n) = p \times n.$$

It therefore suffices to show that their (k, j) entries agree for all $k = 1, \dots, p$ and $j = 1, \dots, n$.

By the definition of the denominator layout, the (k, j) entry of the left-hand side is the partial derivative of the j -th entry of the vector $z(\vec{x})Y(\vec{x})$ with respect to x_k . Applying the ordinary product rule to this scalar,

$$\left(\frac{\partial(zY)}{\partial \vec{x}} \right)_{kj} = \frac{\partial(z(\vec{x}) Y_j(\vec{x}))}{\partial x_k} = z(\vec{x}) \frac{\partial Y_j(\vec{x})}{\partial x_k} + \frac{\partial z(\vec{x})}{\partial x_k} Y_j(\vec{x}).$$

First term. $\partial Y_j / \partial x_k$ is the (k, j) entry of $\frac{\partial Y}{\partial \vec{x}}$, and z is a scalar multiplying every entry, so this is the (k, j) entry of $z \frac{\partial Y}{\partial \vec{x}}$.

Second term. Its row index k comes from $\frac{\partial z}{\partial \vec{x}}$ and its column index j from Y . A matrix whose (k, j) entry has the form $u_k v_j$ is precisely the outer product $\vec{u} \vec{v}^T$; here $\vec{u} = \frac{\partial z}{\partial \vec{x}}$ and $\vec{v} = Y$, so this is the (k, j) entry of $\frac{\partial z}{\partial \vec{x}} Y^T$.

Since the entries agree for every k and j ,

$$\frac{\partial}{\partial \vec{x}}(z(\vec{x}) Y(\vec{x})) = z(\vec{x}) \frac{\partial Y(\vec{x})}{\partial \vec{x}} + \frac{\partial z(\vec{x})}{\partial \vec{x}} Y(\vec{x})^T. \quad \blacksquare$$

Illustration ($p = 2, n = 3$). Applying the product rule entrywise and splitting the result,

$$\frac{\partial(zY)}{\partial \vec{x}} = z \begin{bmatrix} \frac{\partial Y_1}{\partial x_1} & \frac{\partial Y_2}{\partial x_1} & \frac{\partial Y_3}{\partial x_1} \\ \frac{\partial Y_1}{\partial x_2} & \frac{\partial Y_2}{\partial x_2} & \frac{\partial Y_3}{\partial x_2} \end{bmatrix} + \begin{bmatrix} \frac{\partial z}{\partial x_1} Y_1 & \frac{\partial z}{\partial x_1} Y_2 & \frac{\partial z}{\partial x_1} Y_3 \\ \frac{\partial z}{\partial x_2} Y_1 & \frac{\partial z}{\partial x_2} Y_2 & \frac{\partial z}{\partial x_2} Y_3 \end{bmatrix},$$

and the second matrix factors as $\begin{bmatrix} \frac{\partial z}{\partial x_1} \\ \frac{\partial z}{\partial x_2} \end{bmatrix} \begin{bmatrix} Y_1 & Y_2 & Y_3 \end{bmatrix} = \frac{\partial z}{\partial \vec{x}} Y^T$.

Problem 9

Let X be an $n \times d$ constant matrix, $\vec{y}$ a constant vector in $\mathbb{R}^n$, and λ a constant real number. Let

$$J(\vec{\beta}) := (XX^T\vec{\beta} - \vec{y})^T (XX^T\vec{\beta} - \vec{y}) + \lambda \vec{\beta}^T XX^T \vec{\beta}.$$

Find $\frac{\partial}{\partial \vec{\beta}}(J(\vec{\beta}))$, and use $\frac{\partial}{\partial \vec{\beta}}(J(\vec{\beta})) = \vec{0}$ to find the critical points.

Solution

Setup. Let $M := XX^T$, an $n \times n$ matrix, so that $\vec{\beta} \in \mathbb{R}^n$. Note that M is symmetric:

$$M^T = (XX^T)^T = (X^T)^T X^T = XX^T = M.$$

With this notation,

$$J(\vec{\beta}) = (M\vec{\beta} - \vec{y})^T (M\vec{\beta} - \vec{y}) + \lambda \vec{\beta}^T M \vec{\beta}.$$

Expanding. Since $(M\vec{\beta} - \vec{y})^T = \vec{\beta}^T M^T - \vec{y}^T$,

$$(\vec{\beta}^T M^T - \vec{y}^T)(M\vec{\beta} - \vec{y}) = \vec{\beta}^T M^T M \vec{\beta} - \vec{y}^T M \vec{\beta} - \vec{\beta}^T M^T \vec{y} + \vec{y}^T \vec{y}.$$

The two middle terms are 1×1 matrices, hence scalars, and a scalar equals its own transpose: $(\vec{\beta}^T M^T \vec{y})^T = \vec{y}^T M \vec{\beta}$. They therefore combine, giving

$$J(\vec{\beta}) = \underbrace{\vec{\beta}^T M^T M \vec{\beta}}_{\text{quadratic}} - \underbrace{2\vec{y}^T M \vec{\beta}}_{\text{linear}} + \underbrace{\vec{y}^T \vec{y}}_{\text{constant}} + \underbrace{\lambda \vec{\beta}^T M \vec{\beta}}_{\text{quadratic}}.$$

Differentiating term by term.

Quadratic terms. Since $M^T = M$, we have $M^T M = M^2$, and both M^2 and λM are symmetric. Using $\frac{\partial}{\partial \vec{\beta}}(\vec{\beta}^T A \vec{\beta}) = (A + A^T)\vec{\beta} = 2A\vec{\beta}$ for symmetric A ,

$$\frac{\partial}{\partial \vec{\beta}}(\vec{\beta}^T M^T M \vec{\beta}) = 2M^2 \vec{\beta}, \quad \frac{\partial}{\partial \vec{\beta}}(\lambda \vec{\beta}^T M \vec{\beta}) = 2\lambda M \vec{\beta}.$$

Linear term. Write $-2\vec{y}^T M \vec{\beta} = \vec{a}^T \vec{\beta}$ with $\vec{a}^T = -2\vec{y}^T M$. By Problem 6, the derivative is $\vec{a}$, and since $M^T = M$,

$$\frac{\partial}{\partial \vec{\beta}}(-2\vec{y}^T M \vec{\beta}) = (-2\vec{y}^T M)^T = -2M^T \vec{y} = -2M \vec{y}.$$

Constant term. $\frac{\partial}{\partial \vec{\beta}}(\vec{y}^T \vec{y}) = \vec{0}$.

Adding the pieces,

$$\frac{\partial J}{\partial \vec{\beta}} = 2M^2 \vec{\beta} - 2M \vec{y} + 2\lambda M \vec{\beta} = 2(XX^T)^2 \vec{\beta} - 2XX^T \vec{y} + 2\lambda XX^T \vec{\beta}$$

Every term is $n \times 1$, matching the shape of $\vec{\beta}$.

Critical points. Setting the gradient equal to $\vec{0}$ and dividing by 2,

$$M^2\vec{\beta} + \lambda M\vec{\beta} - M\vec{y} = \vec{0}.$$

Factoring M out on the left,

$$M(M\vec{\beta} + \lambda\vec{\beta} - \vec{y}) = \vec{0} \quad \Longleftrightarrow \quad M((M + \lambda I)\vec{\beta} - \vec{y}) = \vec{0}.$$

(The identity matrix is needed because a scalar cannot be added to a matrix.) Hence $\vec{\beta}$ is a critical point precisely when

$$(M + \lambda I)\vec{\beta} - \vec{y} \in \ker M.$$

If $M = XX^T$ is invertible, then $\ker M = \{\vec{0}\}$, so the condition reduces to $(M + \lambda I)\vec{\beta} = \vec{y}$. Multiplying on the left by $(M + \lambda I)^{-1}$ gives the unique critical point

$$\boxed{\vec{\beta} = (M + \lambda I)^{-1}\vec{y} = (XX^T + \lambda I)^{-1}\vec{y}}$$

If M is singular — which occurs whenever $d < n$, since $\text{rank } M \leq d$ — then $\ker M \neq \{\vec{0}\}$ and the critical points form an affine set rather than a single point.